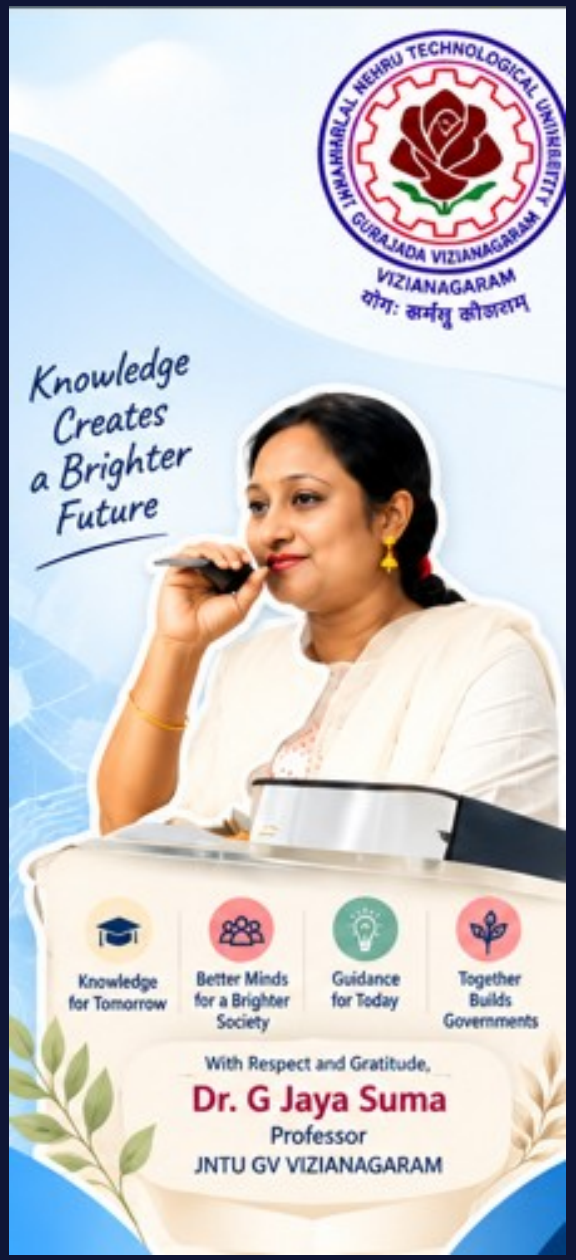


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VIDEO LECTURE SERIES · UNIT I · TOPIC 1

Quantum Computing

Course Introduction & Why Quantum Computing Matters



Superposition

Entanglement

Interference

1

Welcome to the course

A five-unit video lecture series that takes you from qubits to real applications.

5
units

25
hours (5 × 300 min)

43
lecture topics

YOUR INSTRUCTOR

Dr. G Jaya Suma

Professor, JNTU GV Vizianagaram

How every unit works

Learn the idea

Concepts explained with intuition and just the math you need.

→

See it worked

Step-by-step examples with amplitudes, gates and circuits.

→

Run it

Qiskit demos on simulators and the IBM Quantum cloud.

→

Check yourself

Recap, exercises and self-assessment questions.

Goal: understand what quantum computers are, how they compute, and where they genuinely help.

2 What is quantum computing?

Computing that uses the rules of quantum mechanics to store and process information.

In simple words

A classical computer works with bits. A quantum computer works with qubits, whose states can be combined and made to interfere.

Classical bit
Either 0 or 1

→

Qubit
A blend of 0 and 1 set by amplitudes

Three resources quantum algorithms use

Superposition
Combine basis states using amplitudes.

Entanglement
Qubits share correlations no classical system can.

Interference
Amplitudes add or cancel to favor right answers.

Why does it matter?

Exponential state space
n qubits carry 2^n amplitudes, a scale classical machines struggle to track.

Big speedups, select problems
Factoring and quantum simulation are the flagship examples.

New tools, new risks
Better science and optimization, but today's encryption must evolve.

Not a faster laptop — a different kind of machine, useful for specific problems.

3 Roadmap: five units, one connected story

Each unit builds directly on the one before it.

I — 5 hours

Foundations of Quantum Mechanics for Computing

- Qubits, superposition
- Bloch sphere
- Measurement
- Entanglement

You are here

II — 5 hours

Quantum Gates and Circuits

- Reversible, unitary gates
- Pauli, H, T, CNOT
- Circuit model
- Universal gate sets

III — 5 hours

Quantum Algorithms

- Oracles, parallelism
- Deutsch–Jozsa
- Fourier transform
- Grover and Shor

IV — 5 hours

Quantum Programming & Simulation Platforms

- Qiskit fundamentals
- Simulators vs cloud QPUs
- Noise, decoherence
- Error correction

V — 5 hours

Quantum Computing Applications & Emerging Trends

- Quantum cryptography
- Post-quantum crypto
- QML, optimization
- Chemistry, careers

How the units connect

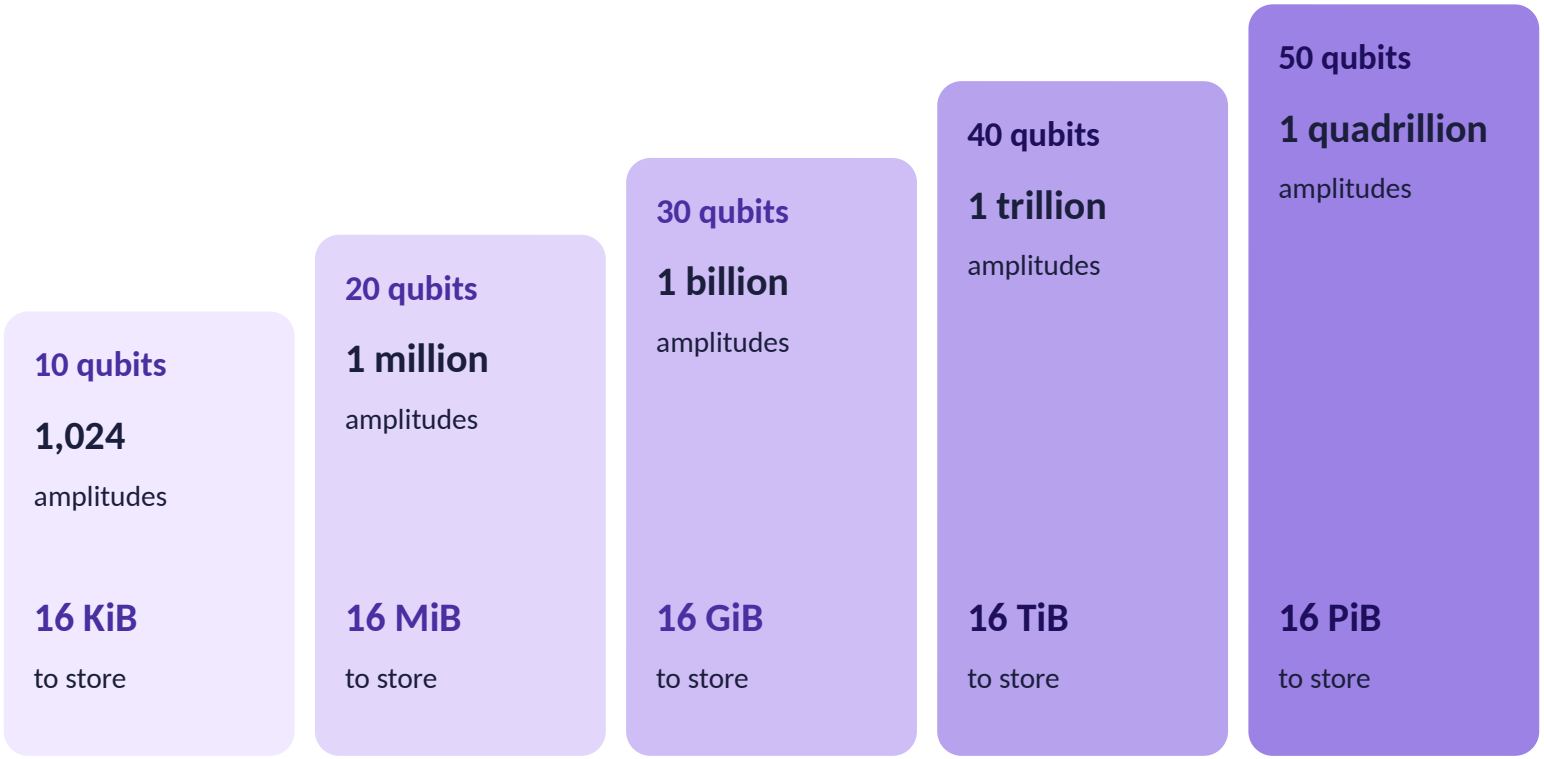
Understand states → Operate on them → Design algorithms → Run on machines → Apply to problems

Units I–IV close with a recap and self-assessment; Unit V closes with course wrap-up and project guidance.

4 Why do classical computers hit a wall?

The state of a quantum system grows exponentially with every qubit added.

Each added qubit doubles the amplitudes to track (bars not to scale)



What the wall really is

1. A state of n qubits needs 2^n complex amplitudes.
2. Storing them classically costs about 16 bytes each, so memory doubles with every qubit.
3. Exact state-vector simulation becomes impractical beyond roughly 50 qubits.

Doubling makes quantum states hard to track classically — it does not mean every problem gets an exponential speedup.

5

Three problems classical computers find hard

Cryptography, chemistry and optimization motivate the field.

Cryptography

THE WALL

RSA security rests on factoring huge numbers, which is infeasible for classical computers at scale.

THE QUANTUM IDEA

Shor's algorithm factors in polynomial time on a large, error-corrected quantum computer.

Example: RSA-2048 is safe today; breaking it needs a machine far beyond current hardware.

Covered in Units III and V

Chemistry

THE WALL

Molecules are quantum systems, and the cost of exact simulation grows exponentially with electrons.

THE QUANTUM IDEA

Feynman's 1981 idea: use a quantum system to simulate quantum systems directly.

Example: Catalysts, batteries and drug-molecule energies.

Covered in Unit V

Optimization

THE WALL

The number of possible routes or schedules explodes as the problem grows.

THE QUANTUM IDEA

Grover search gives a quadratic speedup; annealing and variational methods are still being evaluated.

Example: Delivery routing, scheduling and portfolio selection.

Covered in Units III and V

6

How did quantum computing evolve?

From a theoretical idea to hardware you can use from the cloud.

Ideas and algorithms · 1981–1996

Hardware and standards · 2016–2024

1981

Feynman's proposal

Simulating nature efficiently needs a computer built on quantum mechanics.

1994

Shor's algorithm

Factoring in polynomial time puts RSA-style encryption in the spotlight.

1996

Grover's algorithm

A quadratic speedup for unstructured search.

2016

Cloud access

IBM opens a small quantum processor to the public online.

2019

Advantage claim

Google's 53-qubit sampling experiment claims advantage; later contested.

2024

Error correction and new standards

Google's Willow shows error-correction gains; NIST issues post-quantum standards.

Next: fault-tolerant machines

Many error-corrected logical qubits running long algorithms — the goal of today's hardware roadmaps.

7 Hype vs reality: what can it do today?

Separating solid results from headlines.

What quantum computers can do

- Run algorithms with proven speedups on specific problems, such as factoring and search
- Simulate quantum systems naturally, a big win for chemistry and materials
- Run small circuits on real hardware today through cloud platforms
- Demonstrate error correction on small numbers of qubits

What they cannot do (today)

- Replace laptops or speed up every program
- Try every answer at once and simply read out the best one
- Break RSA-2048 today, which needs a large fault-tolerant machine
- Run long, deep circuits reliably: qubits are still noisy

Rule of thumb: ask which problem, what speedup, on what hardware — and when?

8 What does this course assume?

Linear algebra and discrete mathematics, nothing more.

Linear algebra

- Complex numbers and Euler form
- Vectors, inner products and norms
- Matrices: unitary and Hermitian
- Tensor (Kronecker) products

Discrete mathematics

- Boolean logic and truth tables
- Binary numbers and counting
- Sets, functions and modular arithmetic

Not required

- A physics or quantum mechanics background
- Prior exposure to quantum computing
- Mathematics beyond these two areas

Topic 2 (45 min) refreshes linear algebra up to Hilbert space, and Topic 3 introduces Dirac notation.

9

Where is quantum computing heading?

Every application area maps to a part of this course.

Area	Quantum application	Where you will study it
Security	Quantum key distribution and post-quantum cryptography	Unit V · Topics 2–3
Chemistry and materials	Molecular and materials simulation	Unit V · Topic 6
Optimization	Routing and scheduling with annealing and variational methods	Unit V · Topic 5
Machine learning	Quantum machine learning models	Unit V · Topic 4
Algorithms	Search, factoring and Fourier transforms	Unit III · Topics 4–6
Platforms and hardware	Simulators, cloud QPUs, noise and error correction	Unit IV · Topics 2–7
Industry and careers	Industry landscape and career pathways	Unit V · Topic 7

The big picture

From classical limits to a quantum future

Classical bits	Each bit is 0 or 1: one definite value, processed with Boolean logic.
Qubits	Described by amplitudes; n qubits carry 2^n amplitudes.
The wall	Factoring, molecular simulation and huge search spaces overwhelm classical methods.
The opportunity	Interference-based algorithms give speedups on specific problems.
Reality today	Noisy, small machines; error correction is the road to useful scale.

The course arc

Bits → Qubits → Gates → Algorithms → Applications

One-line takeaway

Quantum computing uses superposition, entanglement and interference to tackle certain problems that classical computers cannot handle efficiently.

Different, not just faster

Specific problems

Still maturing

Up next

Topic 2 · Mathematical Toolkit: Linear Algebra Refresher (45 min)

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QUANTUM
COMPUTING
From Numbers to Qubits

UNIT I · MATHEMATICAL FOUNDATIONS

Topic 1
Mathematical Toolkit: Linear Algebra Refresher

QUANTUM COMPUTING

Unit I
Linear Algebra

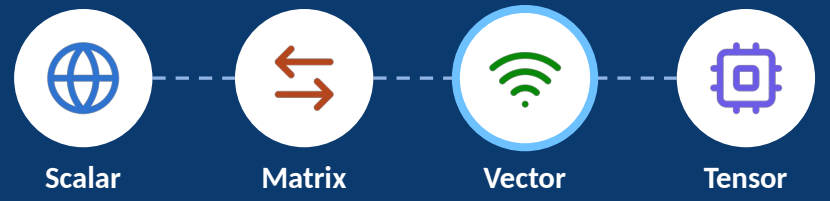
Refresher for quantum computing

Scalars | Matrices | Vectors | Tensors



Vectors vs Related Concepts

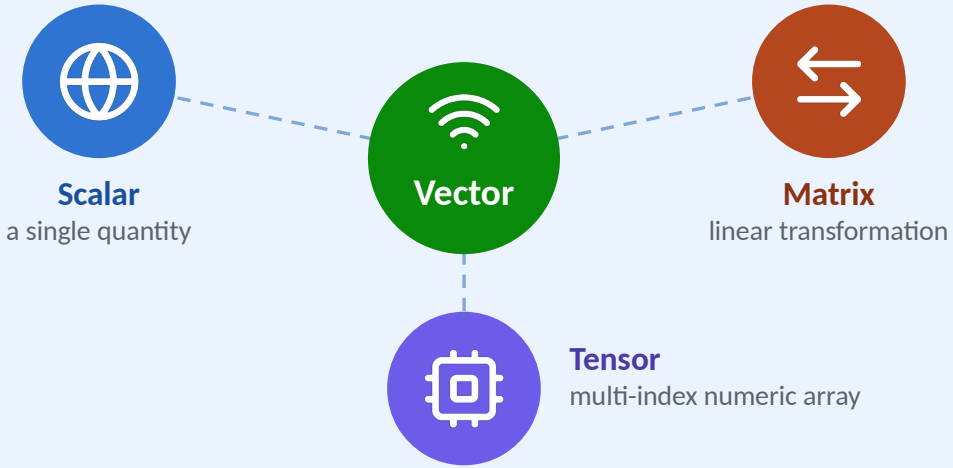
A vector is often confused with a scalar, a matrix and a tensor. They overlap and share numeric structure, but each has a different role.



Related. Overlapping. Not identical.

In Simple Words

A scalar is a single number. A matrix maps one vector to another. A tensor generalises vectors and matrices to higher order. A vector is an ordered list of numbers describing a quantum state.



How Is a Vector Different?

A vector overlaps with each concept, but adds something more:

Vector vs Scalar

A scalar is one number. A vector is an ordered list of numbers — the state of a qubit is always written as a vector.

Vector vs Matrix

A matrix acts directly on one vector. A vector is what gates and operators act upon to change a quantum state.

Vector vs Tensor

A tensor combines multiple vector spaces. A vector is the simplest case — a rank-1 tensor.

A vector overlaps with all three, but it is the one that directly represents the state of a quantum system.

1 Why do these ideas get confused?

They share the same numeric building blocks, but each answers a different question.

1 They share the same building blocks

- Numbers
- Basis
- Operations
- Amplitude
- Index
- Product

Each concept uses some mix of these ingredients.

Example: a two-qubit state is a vector, built as a tensor product, transformed by a matrix and scaled by complex amplitudes — touching all four ideas.

2 But each one asks a different question

Scalar	How do we scale or weight a quantity?
Matrix	How do we transform one vector into another?
Vector	How do we represent a quantum state?
Tensor	How do we combine multiple qubit systems?

Different questions, overlapping mathematics.

2

Vector vs Scalar

A scalar is a single quantity. A vector is an ordered list of quantities.

Scalar

a single number

DIMENSION

One quantity — magnitude only

WRITTEN AS

A plain number, e.g. 0.5

MAIN PURPOSE

Scale or weight another quantity

EXAMPLE

A probability, like 0.5

vs

Vector

an ordered list of numbers

DIMENSION

Many quantities — magnitude and direction

WRITTEN AS

A column of complex amplitudes

MAIN PURPOSE

Describe the full state of a quantum system

EXAMPLE

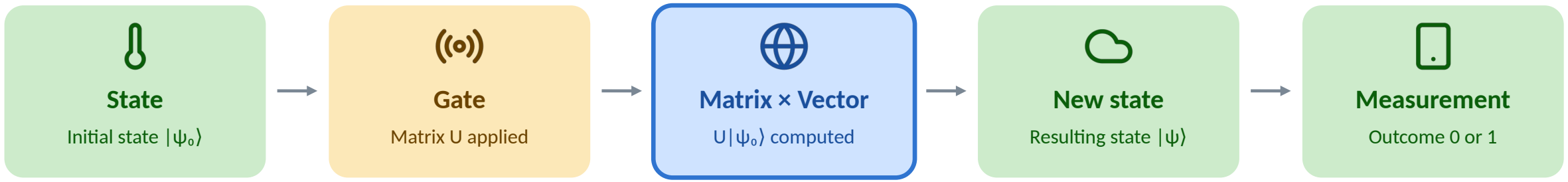
A qubit state $|\psi\rangle = (0.6, 0.8)$

Key idea: a scalar is a single number; a vector is an ordered list of numbers — every quantum state is written as a vector.

2

How a Qubit State Evolves

A quantum gate is a matrix; applying it moves one state vector to another.



Only the middle step multiplies matrix by vector — everything before and after is state and measurement.

Matrices enable evolution

A gate is a matrix; multiplying it by the current state vector produces the new state vector.

Example: a Hadamard gate rotates $|0\rangle$ into an equal superposition.

Vectors go beyond matrices

The vector also encodes complex amplitudes, whose squared magnitudes give measurement probabilities.

Example: measuring $|\psi\rangle = (0.6, 0.8)$ gives outcome 1 with probability 0.64.

Note: not every step is a full unitary matrix — measurement itself is not reversible and collapses the state.

2

Real Numbers vs Complex Numbers

A real number has one part. A complex number has a real part and an imaginary part.

Real Number

ordinary numbers

FORM

A single value on the number line

WRITTEN AS

a, e.g. 0.6

MAIN PURPOSE

Represent classical probabilities directly

EXAMPLE

A classical bit probability, 0.5

vs

Complex Number

numbers with a phase

FORM

A real part and an imaginary part together

WRITTEN AS

a + bi, e.g. 0.6 + 0.8i

MAIN PURPOSE

Carry both magnitude and phase of a quantum amplitude

EXAMPLE

A qubit amplitude, 0.6 + 0.8i

Key idea: a real number has magnitude only; a complex amplitude has magnitude and phase — squaring its magnitude gives a probability (the Born rule).

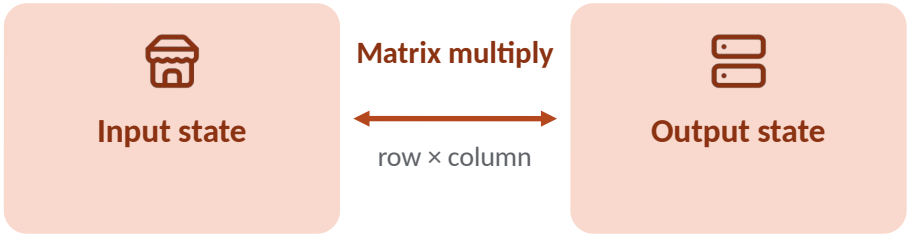
3

Vector vs Matrix

A matrix acts directly on one vector. Many vectors and matrices together build a full quantum circuit.

Matrix

Linear Transformation



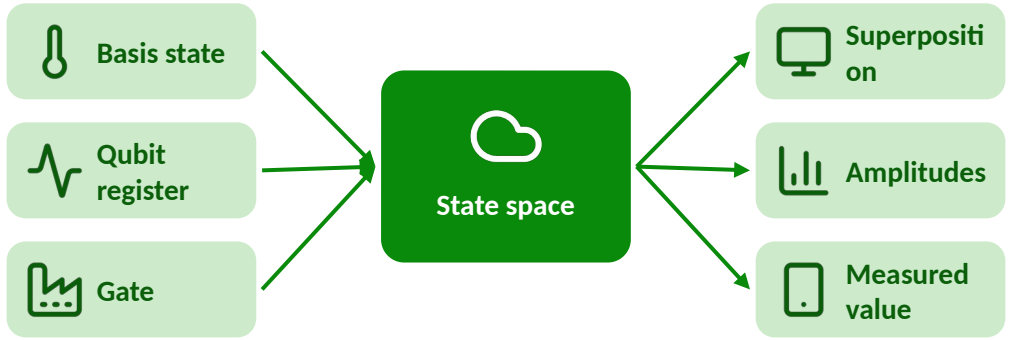
- A matrix acts on one vector, no extra structure needed
- Usually represents one fixed operation, a gate
- Same matrix works on any compatible vector

Example: a NOT gate matrix flips |0⟩ to |1⟩

vs

Vector

Quantum State



- A vector lives in a shared state space with every other state
- Many vectors describe many possible systems
- Vector data feeds measurement and probability

Example: a 3-qubit register is an 8-dimensional state vector

3

Vector vs Matrix: side by side

A matrix is often described as the tool that acts on a vector.

	Matrix	Vector
Structure	Rows and columns of numbers	A single column of numbers
Notation	Capital letter, e.g. U, A	Ket notation, e.g. $\|\psi\rangle$
Role	Transforms or measures a state	Represents the state itself
Size	n × n square (for a gate)	n × 1 column
Key property	Must be unitary to be a valid gate	Must have norm (length) equal to 1
Example	Pauli-X, Hadamard, CNOT	$\|0\rangle$, $\|1\rangle$, $\|+\rangle$

Key idea: a matrix is the operation; a vector is the state it operates on — quantum computing is built from both.

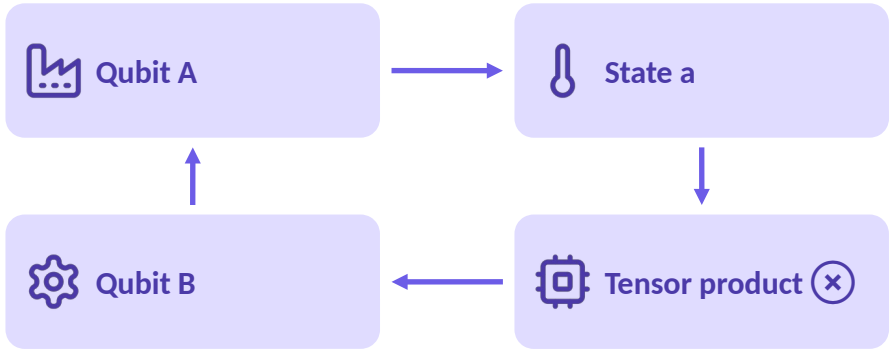
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Vector vs Tensor

A tensor combines multiple vector spaces. A vector is the simplest tensor — a single system on its own.

Tensor

Multi-Qubit System



Combined system

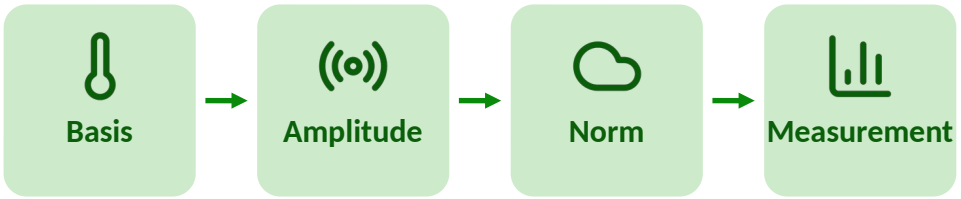
- A tensor product multiplies the dimensions of each subsystem
- Needed whenever two or more qubits interact
- Not every multi-qubit state factors back into single vectors

Example: two entangled qubits form a 4-dimensional joint state vector

vs

Vector

Single Qubit



Dimension grows: 2 qubits combine into a 4-dimensional space

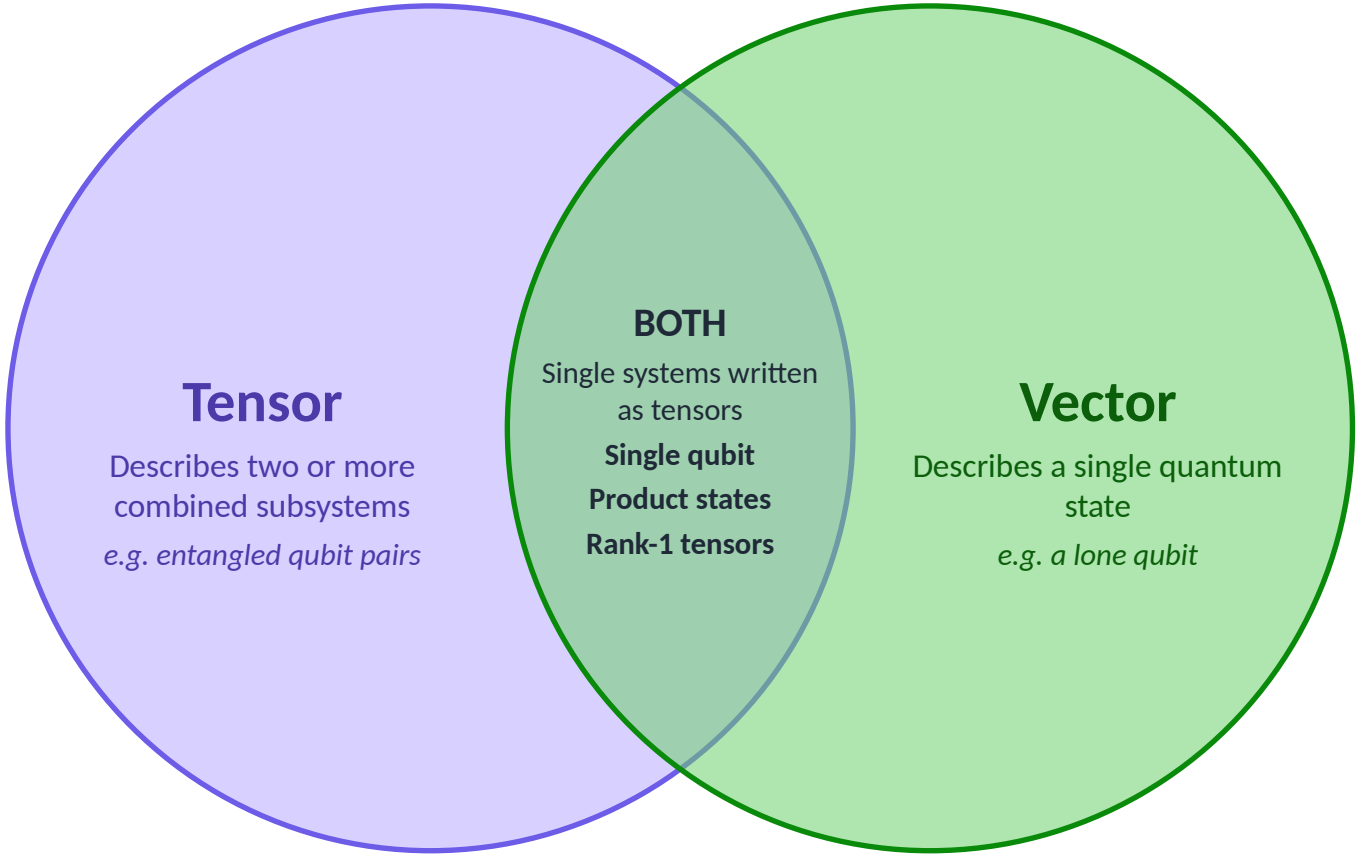
- A vector describes one subsystem on its own
- Focus on a single quantum state and its amplitudes
- Some states are never entangled with another qubit

Example: a single qubit state $|\psi\rangle = (0.6, 0.8)$

4

Vector vs Tensor: where they overlap

Every vector is technically a tensor. What matters is whether the system has one part or many.



A quick test

1. Is there more than one index?
If a quantity needs two or more indices to describe it, it behaves like a higher-rank tensor.

2. Can it be written as one column?
If it can be written as a single column of numbers, it behaves like a vector.

Yes to both?
Then it is a vector that is also a rank-1 tensor — the simplest case of all.

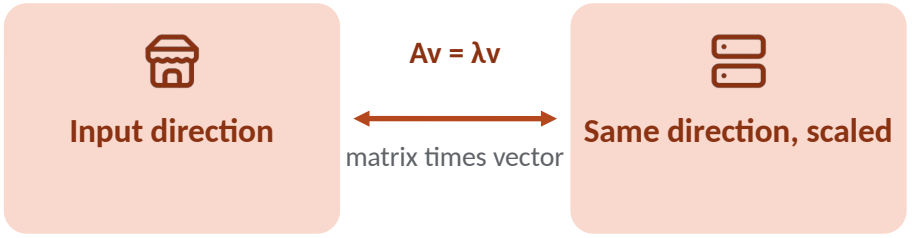
5

Eigenvector vs Eigenvalue

An eigenvector keeps its direction under a matrix. Its eigenvalue is the amount it gets scaled by.

Eigenvalue

Scaling Factor



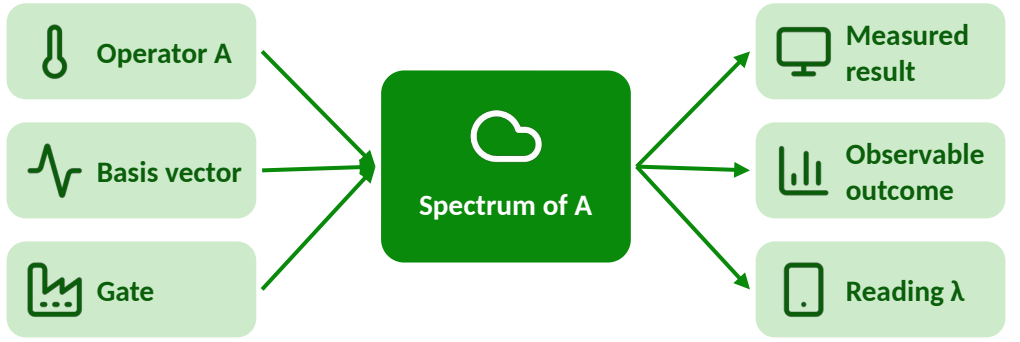
- A number that tells you how much a vector is stretched
- Usually paired one-to-one with its eigenvector
- Real for Hermitian operators, like measurements

Example: measuring spin gives eigenvalues +1 or −1

vs

Eigenvector

Special Direction



- A vector that a matrix scales but never rotates
- Many eigenvectors can form a complete basis
- Used to predict the possible outcomes of measurement

Example: $|0\rangle$ and $|1\rangle$ are eigenvectors of the Z-gate

Comparison at a glance

	Scalar	Matrix	Vector	Tensor
Represents	A single magnitude	A linear transformation	A full quantum state	Multiple combined states
Written as	A plain number	A grid of numbers	A column of numbers	A multi-index array
Main focus	Scaling and weighting	Transforming states (gates)	Describing and measuring states	Combining multiple qubits
Dimension	1 (a point)	$n \times n$	$n \times 1$	Grows with each qubit added
Example	A probability, 0.5	A Hadamard gate	A qubit state $\|\psi\rangle$	Two entangled qubits

The big picture

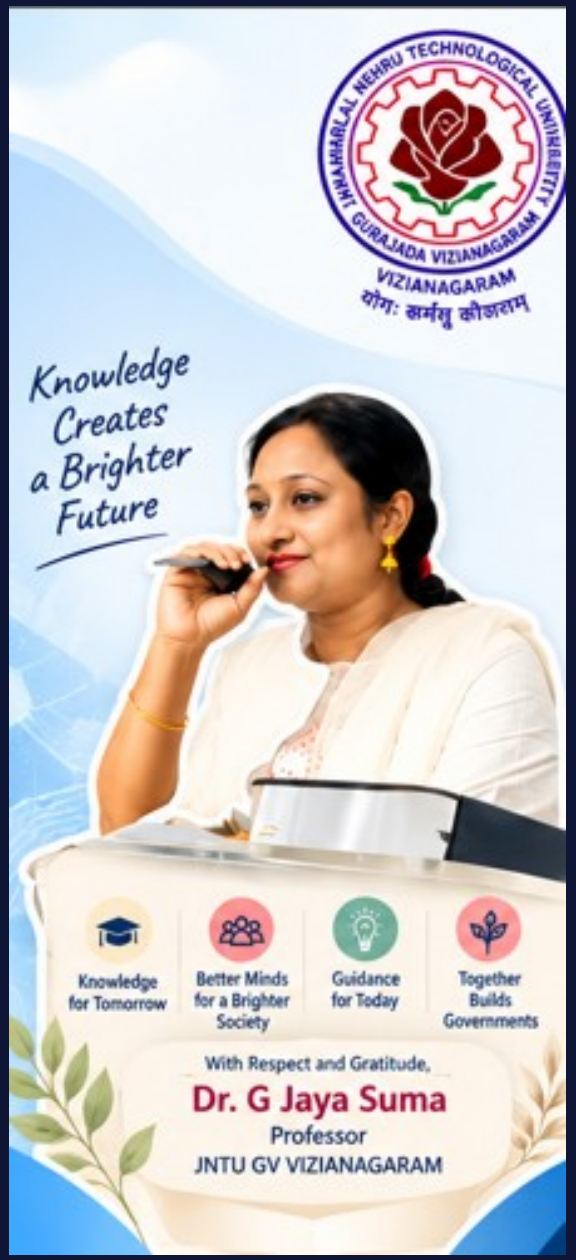
Scalar	A single number that scales or weights.
Matrix	An operator that transforms one state into another.
Vector	The full state of a quantum system, written as a column of amplitudes.
Tensor	Multiple combined states, needed whenever qubits interact.
Remember	Scalar = one number · Matrix = transformation · Vector = state · Tensor = combined systems

One-line takeaway

“Every qubit is a vector, every gate is a matrix, complex amplitudes carry probability, and eigenvalues are what measurement can return.”

State vectors | Gate matrices | Complex amplitudes | Eigen-outcomes

Up next: Qubits, Superposition & Measurement



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VIDEO LECTURE SERIES · UNIT I · TOPIC 2

Classical Bits vs Qubits

How the smallest unit of information changes when quantum mechanics is in charge.

30 MIN

Bit | Qubit | Superposition | Measurement

1 What we'll cover in 30 minutes

Five short segments, building from the familiar bit to the qubit.

1. **Classical bits, recap** — 5 min
   States, determinism, how bits are physically stored.
2. **Meet the qubit** — 6 min
   Dirac notation, amplitudes, and what $|\psi\rangle$ means.
3. **Superposition** — 6 min
   A blend of 0 and 1 — not a coin flip or a mixture.
4. **Bit vs qubit, side by side** — 5 min
   A direct comparison across five properties.
5. **Measurement & scaling** — 8 min
   How collapse works, and why n qubits beat n bits.

2 The classical bit: what you already know

Every digital system, from phones to supercomputers, is built on this one idea.

In simple words

A bit is the smallest unit of classical information. At any instant it holds exactly one of two values — 0 or 1 — and nothing in between.

0 *or* **1**

Physically realized as: voltage high/low, magnetic domain up/down, light pulse present/absent.

= Deterministic

Reading a bit always gives the same, exact value — no randomness involved.

◎ Directly readable

You can inspect a bit at any time without disturbing or destroying it.

• One value at a time

n bits store one specific n-digit pattern — never several patterns together.

3 Meet the qubit

The quantum bit is described by amplitudes, not a single fixed value.

A qubit's state is written

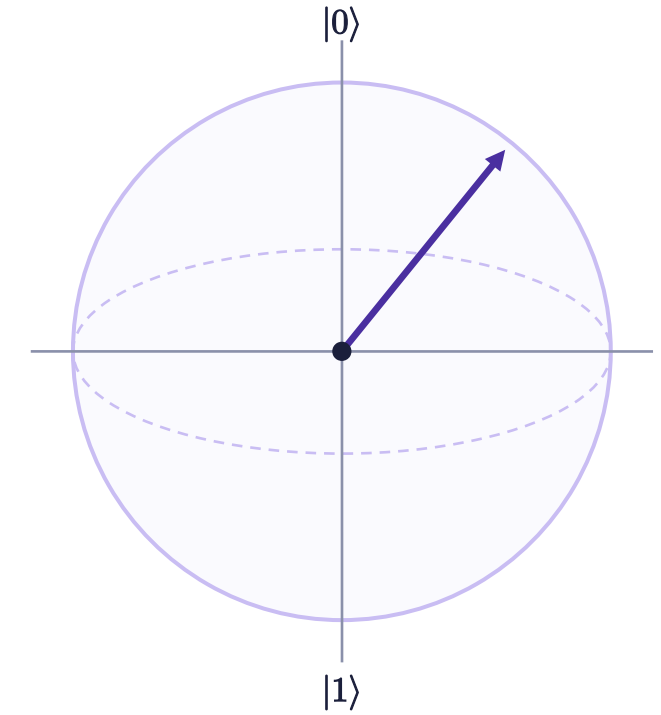
$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

α and β are **amplitudes** — complex numbers that set how much of $|0\rangle$ and $|1\rangle$ are present, with $|\alpha|^2 + |\beta|^2 = 1$.

$|\alpha|^2$ is the probability of measuring 0. $|\beta|^2$ is the probability of measuring 1.

Special cases: α = 1, β = 0 is just $|0\rangle$. α = 0, β = 1 is just $|1\rangle$ — a qubit can still behave exactly like a classical bit.

On the Bloch sphere



A qubit's state is a point (or arrow) on this sphere — $|0\rangle$ and $|1\rangle$ are just its poles.

4

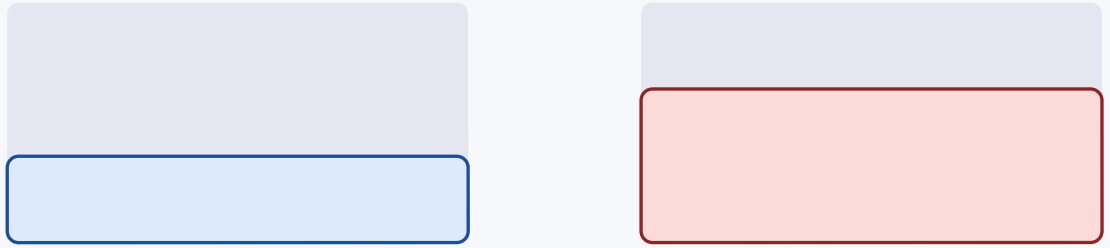
Superposition: a blend, not a coin flip

Until measured, a qubit genuinely holds both amplitudes at once.

Example state

$$|\psi\rangle = 0.6\,|0\rangle + 0.8\,|1\rangle$$

Both amplitudes are non-zero at the same time — the qubit is not "secretly" 0 or 1 while we're not looking.



≠

Not a mixture

A classical mixture (50% chance it's 0, 50% it's 1) has no interference. A superposition does.

±

Amplitudes can cancel

Because amplitudes can be negative or complex, two paths can reinforce or cancel each other — interference.

2^n

Grows with every qubit

n qubits in superposition carry 2^n amplitudes simultaneously, not n independent bits.

5

Bit vs qubit, side by side

Five properties, one direct comparison.

Property	Classical bit	Qubit
Possible states	Exactly one of 0 or 1	Superposition of $\|0\rangle$ and $\|1\rangle$ with amplitudes
Described by	A single value	$\|\psi\rangle = \alpha\|0\rangle + \beta\|1\rangle$
Reading it	Exact, repeatable, no disturbance	Probabilistic — measurement collapses the state
Info in n of them	n bits → 2^n possible patterns, one at a time	n qubits → 2^n amplitudes held at once
Sensitive to noise	Robust — easily error-corrected in hardware	Fragile — decoherence is a central engineering problem

6

Measurement collapses the superposition

Looking at a qubit forces it to pick one classical answer.

Before measurement

$$|\psi\rangle = 0.6\,|0\rangle + 0.8\,|1\rangle$$

Both amplitudes are present. The qubit has not "chosen" a value yet.

↓ *Measure*

After measurement — one outcome

0	1
with prob. 36%	with prob. 64%

Only one of these is ever observed on a single run.

It's irreversible

Once measured, the amplitude information α, β is gone — the qubit is now exactly |0⟩ or |1⟩.

Repeats need repeats

To estimate 36%/64%, run the same circuit many times and count outcomes — one shot never reveals amplitudes.

This is why algorithms use interference

Good quantum algorithms boost the amplitude of the correct answer before that final, one-shot measurement.

7 Why n qubits isn't just n bits

Classical information grows linearly with n; quantum amplitudes grow exponentially.

n bits hold ONE of these patterns at a time · n qubits can hold amplitudes over ALL of them at once

n = 1	n = 2	n = 3	n = 10	n = 20
Classical	Classical	Classical	Classical	Classical
2 patterns	4 patterns	8 patterns	1,024 patterns	~1 million patterns
Quantum	Quantum	Quantum	Quantum	Quantum
2 amplitudes	4 amplitudes	8 amplitudes	1,024 amplitudes	~1 million amplitudes

Same growth pattern you'll see all series: n classical bits give n units of information; n qubits carry 2^n amplitudes to track — the reason exact simulation gets hard fast.

ONE-LINE TAKEAWAY

A classical bit is one fixed value; a qubit is a superposition of amplitudes that only becomes one value when measured.

Deterministic vs probabilistic

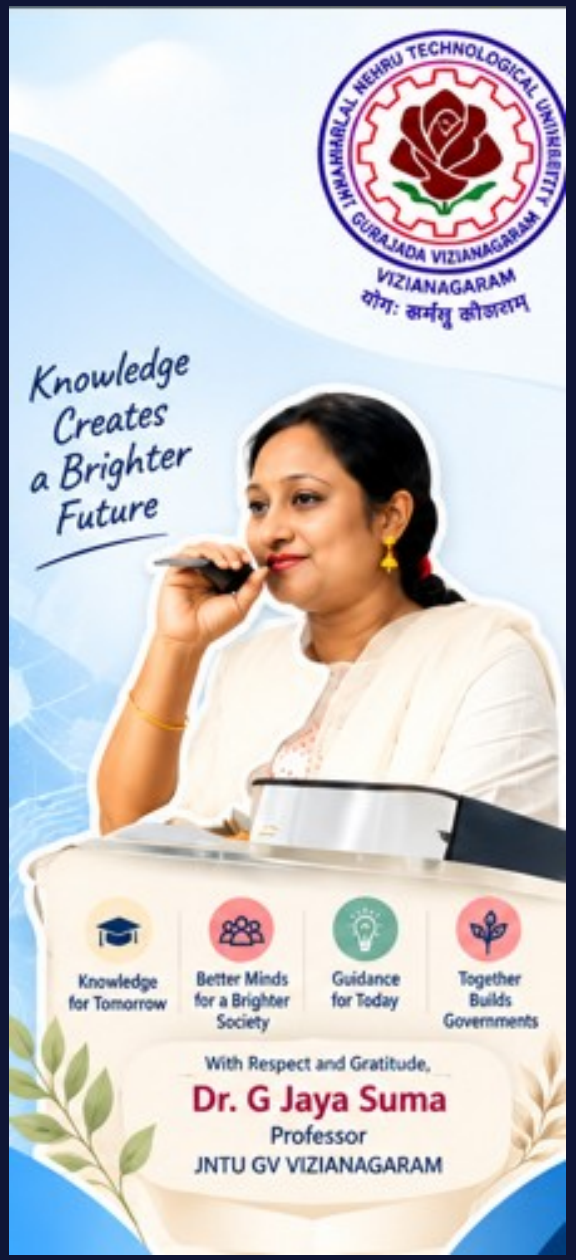
One value vs many amplitudes

Stable vs collapses on reading

→

Next: Topic 3 — Superposition, Entanglement, Interference (35 min)

We build directly on today's amplitudes to introduce the three resources every unit will use.



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VIDEO LECTURE SERIES · UNIT I · TOPIC 3

Superposition in Depth

How a qubit can hold many amplitudes at once — the resource every quantum algorithm depends on.

35 MIN

Blend | Interference | Multi-qubit | Collapse

1 What we'll cover in 35 minutes

Five segments, moving from the qubit recap to a full picture of superposition.

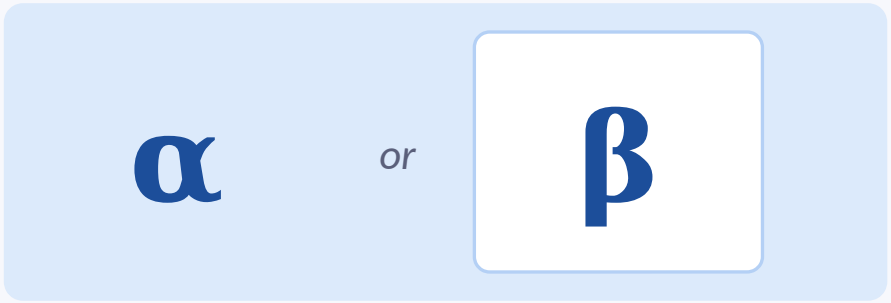
1. **Recap: the qubit's amplitudes** — 5 min
   The notation $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, and what α and β actually mean.
2. **What superposition really is** — 8 min
   Why holding two amplitudes at once is not the same as being undecided.
3. **Superposition vs a mixture** — 7 min
   Why a superposition is fundamentally different from classical uncertainty.
4. **Interference, in and out** — 8 min
   How amplitudes reinforce or cancel — and why that matters.
5. **Multi-qubit superposition & collapse** — 7 min
   Superposition across several qubits, and what happens when we measure.

2 The qubit, recap: two amplitudes, one state

Before going deeper into superposition, let's fix the notation we'll build on.

In simple words

A qubit's state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ is described by two amplitudes, α and β — complex numbers, not probabilities themselves.



Physically realized in photon polarization, electron spin, or superconducting current — any two-level quantum system.

=

Normalized

The amplitudes always satisfy $|\alpha|^2 + |\beta|^2 = 1$ — the probabilities add up to 100%.

◎

Not directly readable

You never see α and β directly — only their squared magnitudes, and only by measuring.

•

Two amplitudes at once

Both $\alpha|0\rangle$ and $\beta|1\rangle$ are present at the same time, until the qubit is measured.

3 What superposition really is

Not two things at once — one state, described by two simultaneous amplitudes.

A superposition is written the same way

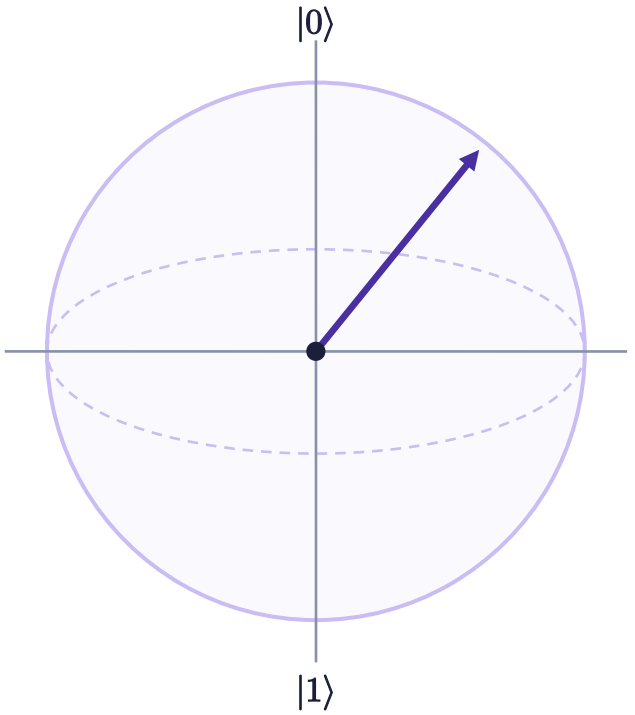
$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

α and β are **both non-zero** when the qubit is genuinely in superposition — it is not secretly $|0\rangle$ or $|1\rangle$ waiting to be revealed.

$|\alpha|^2$ is how much of $|0\rangle$ is really present. $|\beta|^2$ is how much of $|1\rangle$ is really present.

Not superposition: α = 1, β = 0 (or vice versa) puts all the amplitude on one basis state — that's just a bit in disguise, not a blend.

On the Bloch sphere



A true superposition sits away from the poles — anywhere else counts as a blend of $|0\rangle$ and $|1\rangle$.

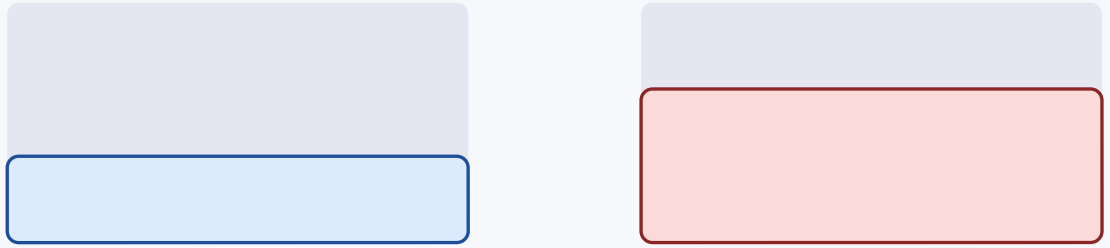
4 Superposition vs. a classical mixture

Until measured, a qubit genuinely holds both amplitudes at once.

Example state

$$|\psi\rangle = 0.6\,|0\rangle + 0.8\,|1\rangle$$

Both amplitudes are non-zero at the same time — the qubit is not "secretly" 0 or 1 while we're not looking.



≠

Not a mixture

A classical mixture (50% chance it's 0, 50% it's 1) has no interference. A superposition does.

±

Amplitudes can cancel

Because amplitudes can be negative or complex, two paths can reinforce or cancel each other — interference.

2^n

Grows with every qubit

n qubits in superposition carry 2^n amplitudes simultaneously, not n independent bits.

5

Interference, in and out

Same amplitudes, two very different outcomes.

Property	Without interference	With interference
Combining two equal paths	Probabilities add: 50% + 50% = 100% chance	Amplitudes add first, then get squared — can exceed or cancel
A negative amplitude	Doesn't exist for probabilities	Subtracts from another path — destructive interference
Two opposite-phase paths	Still sum to some positive chance	Cancel completely — 0% chance for that outcome
Scaling with more paths	More paths simply add more probability	More paths can interfere constructively or destructively
What it enables	Ordinary randomized algorithms	Boosting the correct answer's amplitude before measurement

6

Measurement collapses the superposition

Looking at a qubit forces it to pick one classical answer.

Before measurement

$$|\psi\rangle = 0.6\,|0\rangle + 0.8\,|1\rangle$$

Both amplitudes are present. The qubit has not "chosen" a value yet.

↓ *Measure*

After measurement — one outcome

0	1
with prob. 36%	with prob. 64%

Only one of these is ever observed on a single run.

It's irreversible

Once measured, the amplitude information α, β is gone — the qubit is now exactly |0⟩ or |1⟩.

Repeats need repeats

To estimate 36%/64%, run the same circuit many times and count outcomes — one shot never reveals amplitudes.

This is why algorithms use interference

Good quantum algorithms boost the amplitude of the correct answer before that final, one-shot measurement.

7

Multi-qubit superposition: it's not just n qubits

Put n qubits into superposition together, and the amplitudes to track explode.

n classical bits hold ONE pattern at a time · n qubits in superposition hold amplitudes over ALL of them at once

n = 1	n = 2	n = 3	n = 10	n = 20
Classical	Classical	Classical	Classical	Classical
2 patterns	4 patterns	8 patterns	1,024 patterns	~1 million patterns
Quantum	Quantum	Quantum	Quantum	Quantum
2 amplitudes	4 amplitudes	8 amplitudes	1,024 amplitudes	~1 million amplitudes

The takeaway for this segment: A single qubit gave us 2 amplitudes (α, β). n qubits in joint superposition carry 2^n amplitudes — this is why simulating quantum systems classically gets hard fast.

ONE-LINE TAKEAWAY

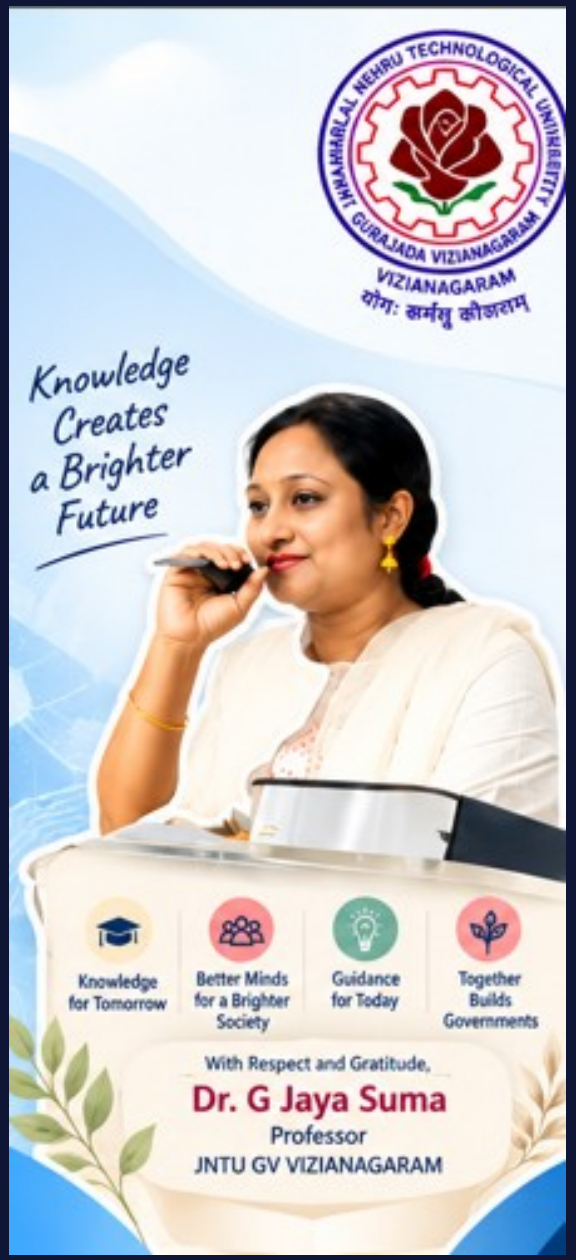
A superposition is one state carrying several amplitudes at once — and interference between them is the resource quantum algorithms exploit.

- One state, many amplitudes
- Amplitudes interfere, not just add
- Collapses to one outcome on measurement

→

Next: Topic 4 — Entanglement

We build directly on today's amplitudes to see what happens when two qubits share a single superposition.



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The Bloch Sphere Representation

Turning a qubit's two amplitudes into a single point on a sphere — the geometric picture behind every single-qubit gate.

40 MIN

Angles | Geometry | Rotations | Phase

1 What we'll cover in 40 minutes

Five segments, moving from the qubit's amplitudes to a full geometric picture on the sphere.

1. **Recap: from amplitudes to a vector** — 5 min
   The notation $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, and why it pins down a single point once we fix a length.
2. **Parametrizing a qubit: θ and φ** — 8 min
   Rewriting α and β as two real angles — a latitude and a longitude.
3. **Mapping the sphere** — 7 min
   Poles, the equator, and why only relative phase is visible.
4. **Special states and their coordinates** — 8 min
   Where $|0\rangle$, $|1\rangle$, $|+\rangle$, $|-\rangle$ and other landmarks sit on the sphere.
5. **Rotations & the limits of the picture** — 12 min
   How gates spin the Bloch vector, and why the sphere stops at one qubit.

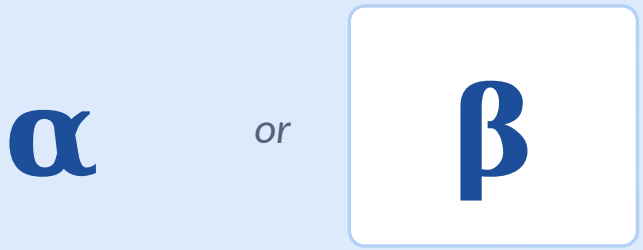
2

The qubit, recap: two amplitudes, one state

Before mapping the sphere, let's fix the notation we'll build the geometry on.

In simple words

A qubit's state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ is described by two amplitudes, α and β — complex numbers, not probabilities themselves.



Physically realized in photon polarization, electron spin, or superconducting current — any two-level quantum system.

=

Normalized

The amplitudes always satisfy $|\alpha|^2 + |\beta|^2 = 1$ — the probabilities add up to 100%.

◎

Not directly readable

You never see α and β directly — only their squared magnitudes, and only by measuring.

•

Two amplitudes at once

Both $\alpha|0\rangle$ and $\beta|1\rangle$ are present at the same time, until the qubit is measured.

3 Parametrizing a qubit: the angles θ and φ

Two complex amplitudes, one length constraint, one invisible phase — exactly two real angles remain.

The angle form of a qubit's state

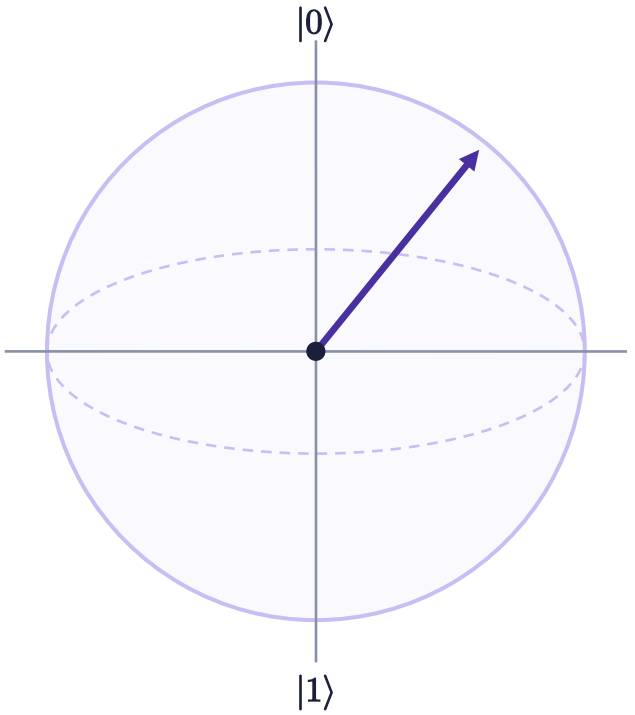
$$|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\varphi}\sin(\theta/2)|1\rangle$$

θ is the **polar angle** — how far the state sits from the $|0\rangle$ pole, running from 0 to π.

φ is the azimuthal angle **— the relative phase** between α and β, running from 0 to 2π.

Same amplitudes, new picture: α = cos(θ/2) and β = e^{iφ}sin(θ/2) are just the old amplitudes written in polar form.

On the Bloch sphere



θ sets the latitude, φ sets the longitude — together they pin down one exact point on the surface.

4

Mapping the sphere: poles and the equator

Two special latitudes, and the circle where every point is a genuine superposition.

Landmark angles

$$\theta = 0 \quad \text{and} \quad \theta = \pi$$

These are the only two points on the sphere with zero superposition — pure $|0\rangle$ and pure $|1\rangle$.

$|0\rangle$

North pole

$|1\rangle$

South pole

◎

The poles

$\theta = 0$ gives $|0\rangle$, $\theta = \pi$ gives $|1\rangle$ — the only points with no superposition at all.

±

The equator

$\theta = \pi/2$ is the circle of maximal superposition — every point on it is an equal mix of $|0\rangle$ and $|1\rangle$.

φ

Longitude = phase

Moving around the equator changes only φ, the relative phase between α and β — never the measurement odds.

5 Special states and their coordinates

The most-used states are just landmark points, easy to locate once you know θ and φ.

State	Angles (θ, φ)	Where it sits
$\|0\rangle$	$\theta = 0$	North pole
$\|1\rangle$	$\theta = \pi$	South pole
$\|+\rangle = (\|0\rangle+\|1\rangle)/\sqrt{2}$	$\theta = \pi/2, \varphi = 0$	Equator, +x axis
$\|-\rangle = (\|0\rangle-\|1\rangle)/\sqrt{2}$	$\theta = \pi/2, \varphi = \pi$	Equator, −x axis
Any pure state	any (θ, φ)	Exactly one point on the surface — never inside

6 Rotations move the Bloch vector

Every single-qubit gate turns out to be a rotation of the point on the sphere.

Before the gate

$|\psi\rangle$ at angles (θ, φ)

The state sits at some point on the sphere's surface, fixed by its two angles.

↓ *Apply a gate*

After the gate — a new point

X gate

a π rotation about the x-axis — flips $|0\rangle$ and $|1\rangle$

Z gate

a rotation about the z-axis — leaves $|0\rangle$, $|1\rangle$ fixed, flips the phase

Only the rotation's axis and angle change from gate to gate.

It's reversible

Unlike measurement, a rotation never destroys information — apply the inverse rotation and the original point comes back exactly.

Every gate is geometric

X, Y, Z, Hadamard, and every phase gate are all just rotations by a fixed angle about a fixed axis.

This is why the sphere matters

Reasoning about a single-qubit algorithm becomes reasoning about where a sequence of rotations sends a point.

7 Where the Bloch-sphere picture breaks down

One sphere per qubit works only because a single qubit needs no more than that.

Naïve idea: one Bloch sphere per qubit · Reality: the real parameters the joint state actually needs

n = 1	n = 2	n = 3	n = 10	n = 20
n spheres	n spheres	n spheres	n spheres	n spheres
2 parameters	4 parameters	6 parameters	20 parameters	40 parameters
Needed	Needed	Needed	Needed	Needed
2 parameters	6 parameters	14 parameters	2,046 parameters	~2.1 million parameters

The takeaway for this segment: A single qubit's state really does fit on one sphere — two real angles, nothing lost. Past one qubit, entanglement means the joint state needs far more parameters than n separate spheres could ever hold.

ONE-LINE TAKEAWAY

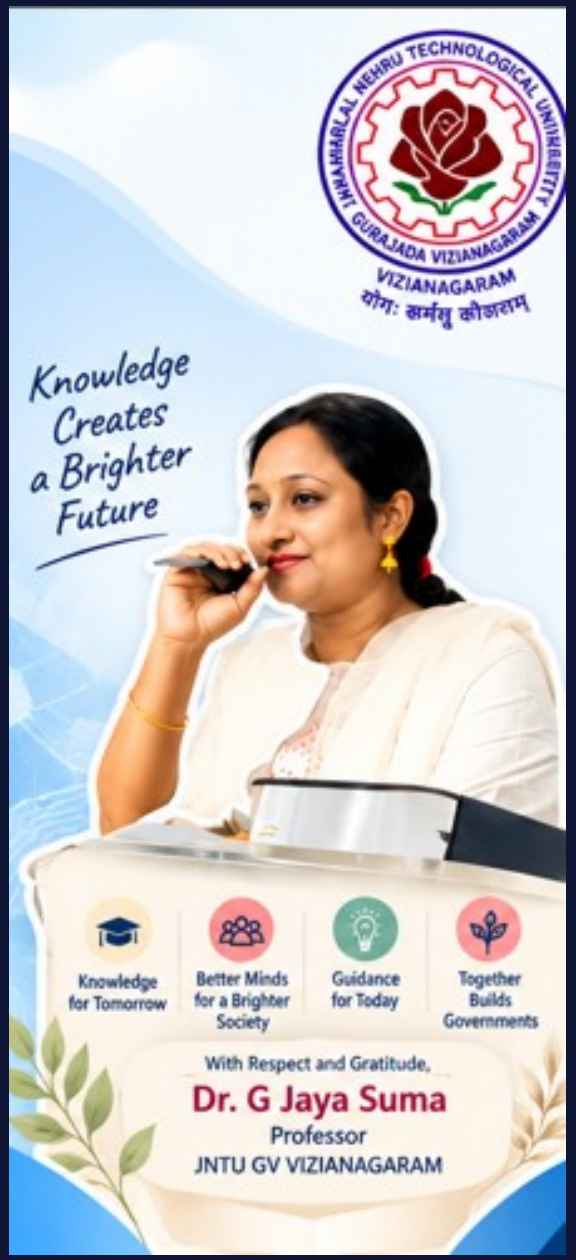
Every pure single-qubit state is a point on a sphere — its two angles are the whole story, and every gate just rotates that point.

- One state, one point on the sphere
- Two angles (θ, φ) — nothing more, nothing less
- Measurement projects it onto a pole

→

Next: Topic 5 — Multi-Qubit States & Entanglement

We leave the single sphere behind to see how two qubits can share one joint state.



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Measurement Postulates

How a quantum state turns into a classical result — the rules behind every quantum readout.

40 MIN

Basis | Born rule | Collapse | Multi-qubit

1 What we'll cover in 40 minutes

Five segments, from what a measurement is to reading out several qubits at once.

1. **Recap: bases and outcomes** — 5 min
   What it means to measure in a basis, and what results we can get.
2. **The measurement postulate (Born rule)** — 9 min
   How measurement operators turn amplitudes into probabilities.
3. **Worked example: amplitudes to numbers** — 8 min
   Applying the Born rule to one concrete qubit state, step by step.
4. **Same state, different basis** — 9 min
   Why the basis we choose changes what a measurement can tell us.
5. **Collapse, repeated shots & multi-qubit readout** — 9 min
   What measuring does to the state, and how to read out several qubits.

2 Recap: a basis, and its possible outcomes

Before the postulate itself, let's fix what we mean by a measurement basis.

In simple words

To measure a qubit, we choose a basis — a pair of reference states such as $|0\rangle$ and $|1\rangle$. Each possible outcome belongs to one basis state.

$|0\rangle$ *or* $|1\rangle$

In hardware the basis is set by the device — photon polarization, electron spin or a superconducting current.

Σ

Complete

Every outcome has its own measurement operator, and together they cover all cases — the probabilities add up to 100%.

◎

Basis-dependent

The same state gives different statistics when measured in a different basis — the choice of basis is ours.

0/1

Classical outcomes

Each measurement returns an ordinary classical result — a definite 0 or 1 — never a blend of the two.

3 The measurement postulate (Born rule)

Amplitudes stay hidden; the postulate gives the probability of each outcome we can see.

The probability of outcome m

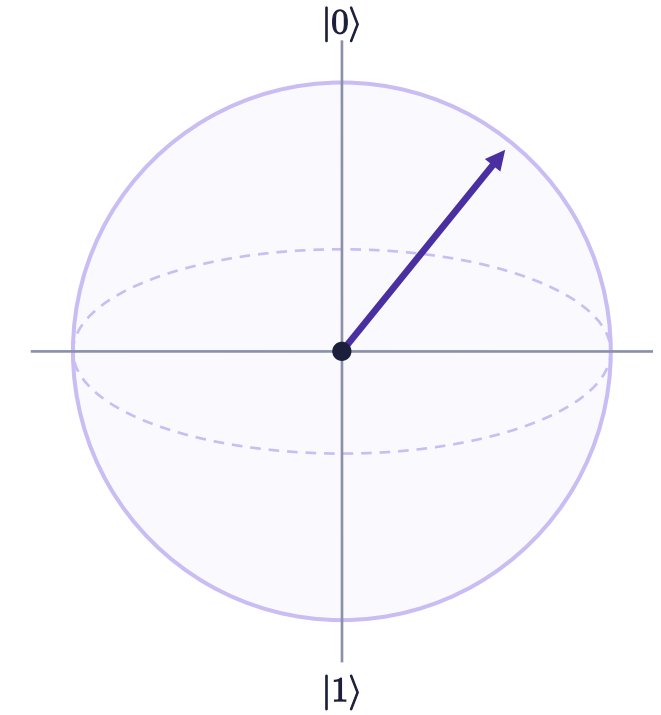
$$p(m) = \langle\psi|M_m{\dagger}M_m|\psi\rangle$$

Each outcome m has its own **measurement operator M_m**; the formula gives the chance of seeing it, and the state then collapses to $M_m|\psi\rangle$, rescaled to length 1.

$M_0 = |0\rangle\langle 0|$ and **$M_1 = |1\rangle\langle 1|$** give $p(0) = |\alpha|^2$ and $p(1) = |\beta|^2$.

Key idea: probabilities come from squared amplitudes — α and β themselves are never observed directly.

On the Bloch sphere



Measuring along the z-axis asks: which pole is closer? The state lands on |0⟩ or |1⟩, with odds set by its tilt.

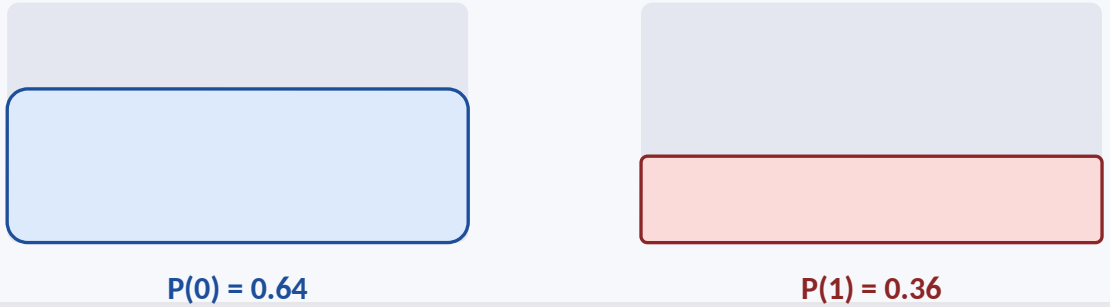
4 Worked example: amplitudes to probabilities

Apply the Born rule to a concrete state — square each amplitude to get its probability.

Example state

$$|\psi\rangle = 0.8\,|0\rangle - 0.6\,|1\rangle$$

The amplitude on $|1\rangle$ is negative, but that doesn't matter here: squaring gives p(0) = 0.64 and p(1) = 0.36.



Σ

Probabilities sum to 1

0.64 + 0.36 = 1. The rule guarantees this whenever the state is normalized.

α^2

Squares hide the sign

The amplitude −0.6 gives the same 36% as +0.6 — sign only matters once we change basis or let paths interfere.

φ

Global phase is invisible

Multiplying the whole state by a phase factor changes no probability — only relative phase between amplitudes counts.

5 Same state, different basis

Measure in the computational basis or the Hadamard basis, and the story changes.

Property	Computational basis	Hadamard basis
Basis states	$\|0\rangle$ and $\|1\rangle$	$\|+\rangle = (\|0\rangle + \|1\rangle)/\sqrt{2}$ and $\|-\rangle = (\|0\rangle - \|1\rangle)/\sqrt{2}$
Outcome operators	$\|0\rangle\langle 0\|$ and $\|1\rangle\langle 1\|$	$\|+\rangle\langle +\|$ and $\|-\rangle\langle -\|$
Measuring $\|0\rangle$ gives	0 with 100% certainty	+ or − with 50% each
Measuring $\|+\rangle$ gives	0 or 1 with 50% each	+ with 100% certainty
What the basis reveals	The magnitudes $\|\alpha\|^2$ and $\|\beta\|^2$	The relative sign (phase) of α and β

6 Measurement collapses the state

One run gives one classical answer — and leaves the qubit in the matching basis state.

Before measurement

$$|\psi\rangle = 0.8\,|0\rangle - 0.6\,|1\rangle$$

The amplitudes fix the odds, but no outcome exists until we measure.

↓ *Measure*

After measurement — one outcome

$\|0\rangle$	$\|1\rangle$
with prob. 64%	with prob. 36%

Only one of these is ever observed on a single run.

Measure twice, get the same answer

Measure again right away and the outcome repeats with certainty — the state is already $|0\rangle$ or $|1\rangle$.

Repeated runs reveal the odds

To estimate 64%/36%, prepare the same state many times and count outcomes — one shot never reveals amplitudes.

The amplitudes are lost

Collapse erases α, β and their relative phase — so algorithms use interference before that final measurement.

7 Multi-qubit measurement: n bits out

Many amplitudes go in, but one measurement returns just a single n-bit string.

n qubits are described by 2^n amplitudes · measuring all n qubits returns only n classical bits

n = 1	n = 2	n = 3	n = 10	n = 20
State holds	State holds	State holds	State holds	State holds
2 amplitudes	4 amplitudes	8 amplitudes	1,024 amplitudes	~1 million amplitudes
One shot returns	One shot returns	One shot returns	One shot returns	One shot returns
1 bit	2 bits	3 bits	10 bits	20 bits

The takeaway for this segment: Measuring all n qubits returns one n-bit string, drawn with probability $|\text{amplitude}|^2$. Measuring only some qubits collapses just the matching part of the state.

ONE-LINE TAKEAWAY

Measurement turns amplitudes into probabilities (the Born rule) and collapses the state to the one outcome we observe.

Probability = $|\text{amplitude}|^2$

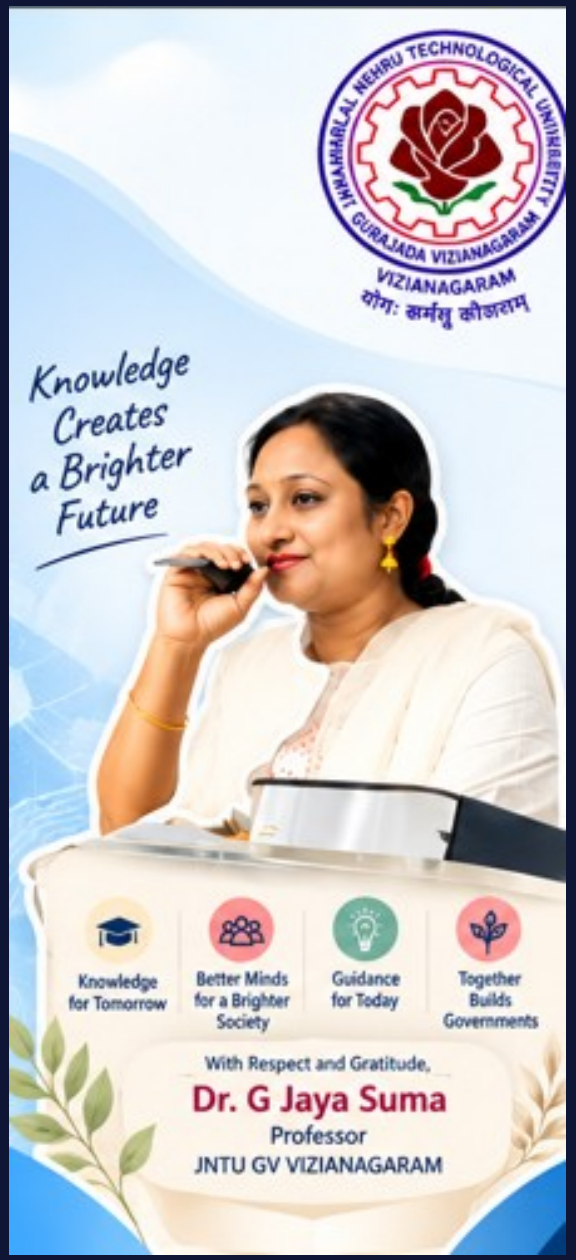
The basis we pick matters

Collapse leaves one outcome

→

Next: Topic 6 — Quantum Gates

We take today's measurement rules and see how gates reshape amplitudes before that final readout.



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Multi-Qubit Systems & Entanglement

How two or more qubits combine into one joint state — and why that state can hold correlations no single qubit ever could.

50 MIN

Tensor | Bell States | Correlation | Scaling

1 What we'll cover in 50 minutes

Five segments, moving from a single qubit to joint states no single qubit can describe alone.

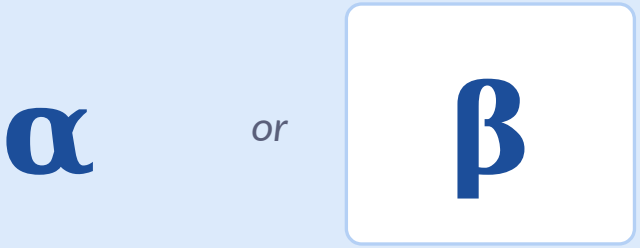
1. **Recap: from one qubit to two** — 5 min
   Each qubit still lives on its own Bloch sphere — until we ask what it means to combine them.
2. **Combining qubits: the tensor product** — 10 min
   How two 2-dimensional state spaces combine into one 4-dimensional space.
3. **Product states vs. entangled states** — 8 min
   Some joint states factor into two separate qubits — and some flatly refuse to.
4. **The Bell states** — 9 min
   Four maximally entangled states that anchor every two-qubit protocol.
5. **Measuring entanglement & why it scales** — 18 min
   What correlated measurement outcomes look like, and how fast the state space grows.

2 One qubit, recap: a point on a sphere

Before combining qubits, let's fix what a single qubit's state already gives us.

In simple words

A single qubit's state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ corresponds to exactly one point on its own Bloch sphere — two real angles, nothing more.



Each qubit — photon, electron spin, superconducting circuit — carries its own independent sphere like this.

=

Normalized

Every valid qubit state still satisfies $|\alpha|^2 + |\beta|^2 = 1$, exactly as before.

◎

One sphere, one qubit

A single Bloch sphere fully captures one qubit — but says nothing about how two qubits relate to each other.

•

The open question

What happens when we have two qubits together — do they still just live on two separate spheres?

3 Combining qubits: the tensor product

Two 2-dimensional spaces combine into one 4-dimensional space — not by adding, but by multiplying.

The joint state of two qubits

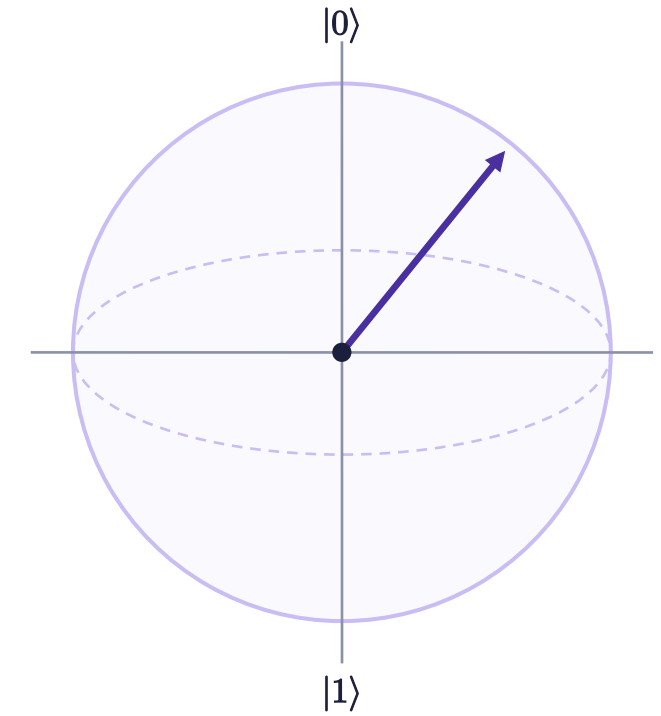
$$|\psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle$$

Four amplitudes **replace two pairs** — one qubit's 2 amplitudes times another's 2 amplitudes gives 4, not 2 + 2.

$|a|^2$ is the chance of measuring 00. $|d|^2$ is the chance of measuring 11.

Not just independent: a, b, c, d can mix the four outcomes in ways that never split into 'qubit 1's amplitudes' times 'qubit 2's amplitudes'.

Each qubit, on its own sphere



This picture is the whole story only when the two qubits aren't entangled — next, we see when it breaks.

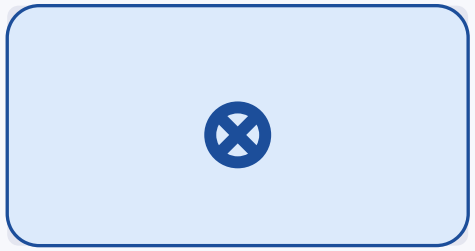
4 Product states vs. entangled states

Some joint states are just two qubits side by side — others fuse into one indivisible whole.

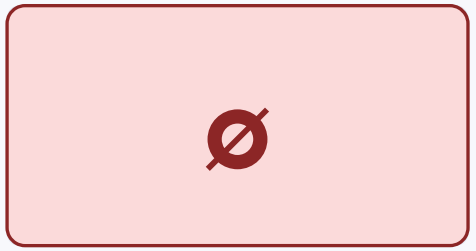
Two joint states, side by side

$$|0\rangle \otimes |+\rangle \quad \text{vs} \quad 0.71|00\rangle + 0.71|11\rangle$$

The left state factors into 'qubit 1 is $|0\rangle$' and 'qubit 2 is $|+\rangle$'. The right one provably does not factor at all.



Product state



Entangled state

Separable

A product state can always be written as (qubit 1's state) ⊗ (qubit 2's state) — two independent descriptions.

Can't factor

No matter how you split the labels, an entangled state can't be written as one qubit's amplitudes times the other's.

No hidden description

There's no secret local value for each qubit that reproduces the correlations — this is the heart of Bell's theorem.

5 The Bell states

Four specific entangled states that anchor teleportation, superdense coding, and every entanglement protocol.

Bell state	Formula	What it means
$\|\Phi+\rangle$	$(\|00\rangle + \|11\rangle)/\sqrt{2}$	Both qubits always agree — measured 00 or 11, never mixed
$\|\Phi-\rangle$	$(\|00\rangle - \|11\rangle)/\sqrt{2}$	Same agreement, with an opposite relative phase
$\|\Psi+\rangle$	$(\|01\rangle + \|10\rangle)/\sqrt{2}$	Both qubits always disagree — measured 01 or 10, never matched
$\|\Psi-\rangle$	$(\|01\rangle - \|10\rangle)/\sqrt{2}$	Same disagreement, with an opposite relative phase
Any Bell state	maximally entangled	Measuring one qubit instantly fixes the other's outcome

6 Measuring one qubit fixes the other

For an entangled pair, a measurement on one qubit instantly determines what the other will show.

Before measurement

$$|\Phi+\rangle = 0.71\ |00\rangle + 0.71\ |11\rangle$$

Neither qubit has a definite value yet — only the pair, together, is fully described.

↓ *Measure qubit 1*

After measurement — qubit 2 is now fixed

00

if qubit 1 reads 0, prob. 50%

11

if qubit 1 reads 1, prob. 50%

Only one of these two outcomes is ever observed on a single run.

It's not signaling

No information travels faster than light — the correlation only shows up once the two results are compared.

Repeats need repeats

To see the 50/50 correlation, repeat the experiment many times and compare the two qubits' outcomes.

This is why entanglement is a resource

Protocols like teleportation and superdense coding are built directly on this shared correlation.

7 Why multi-qubit systems explode in size

Every qubit you add doesn't add to the state space — it multiplies it.

n classical bits hold ONE pattern at a time · n qubits hold amplitudes over ALL of them at once

n = 1	n = 2	n = 3	n = 10	n = 20
Classical	Classical	Classical	Classical	Classical
2 patterns	4 patterns	8 patterns	1,024 patterns	~1 million patterns
Quantum	Quantum	Quantum	Quantum	Quantum
2 amplitudes	4 amplitudes	8 amplitudes	1,024 amplitudes	~1 million amplitudes

The takeaway for this segment: Entanglement means an n-qubit state generally needs all 2^n amplitudes at once — it almost never breaks apart into n separate one-qubit stories, which is exactly why classical computers struggle to simulate it.

ONE-LINE TAKEAWAY

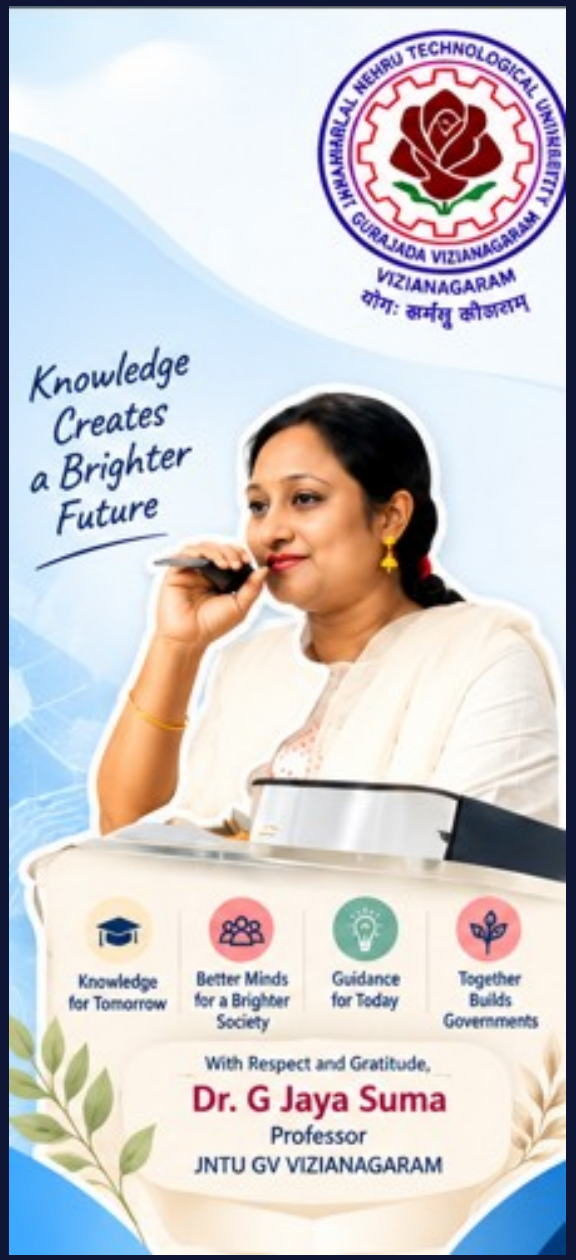
Combining qubits multiplies their state spaces — and entangled states are joint descriptions that no separate, per-qubit story can ever capture.

- Joint states multiply, not add
- Entangled states can't be factored
- One measurement correlates them both

→

Next: Topic 6 — Quantum Gates & Circuits

We start building circuits — sequences of gates that create and exploit entanglement on purpose.



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VIDEO LECTURE SERIES · UNIT I · HANDS-ON SESSION

First Steps with a Quantum Simulator

Build, run and read your first quantum circuits — no quantum hardware needed.

30 MIN

Circuit | Shots | Counts | Bell pair

Fundamentals of Quantum Computing · Hands-On Session

1 What we'll cover in 30 minutes

Five segments, from installing the simulator to running a Bell pair.

1 Set up the simulator

Install Qiskit, import two tools, and learn the build-run-read loop.

4 min

2 Your first circuit

One qubit, one measurement, and the simplest possible result.

6 min

3 Superposition: run it, read the counts

Add an H gate and watch shots turn into probabilities.

7 min

4 Predict, run, compare

Check five small circuits against what theory says.

6 min

5 Two qubits & simulator limits

Build a Bell pair, then see why memory limits simulators.

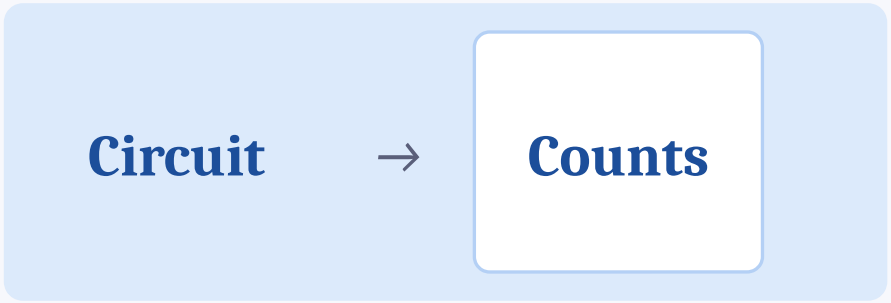
7 min

2 Setup: what a simulator does

A simulator runs quantum circuits on an ordinary computer, so we can experiment without hardware.

In simple words

A quantum simulator is a program that tracks a small circuit's amplitudes on your own computer. You build a circuit, run it many times, and read the counts.



Every experiment today follows the same loop: build a circuit, run it for many shots, read the counts.

1 Install once

In a terminal or notebook cell, run: pip install qiskit qiskit-aer

2 Import the tools

QuantumCircuit and transpile come from qiskit; AerSimulator comes from qiskit_aer.

3 Choose your shots

One shot is one full run ending in a measurement. We'll use 1024 shots per experiment.

3 Your first circuit: one qubit, measured

A few lines of code — and the simplest possible result.

The whole program

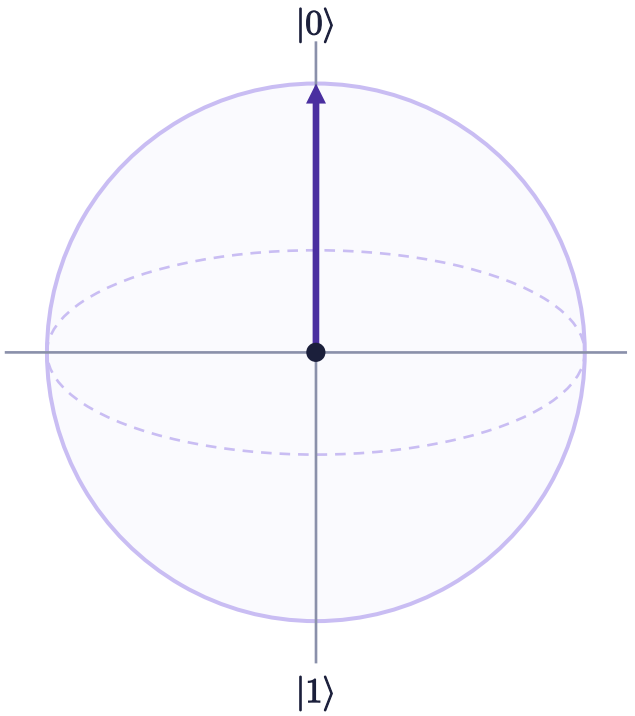
```
sim = AerSimulator()
qc = QuantumCircuit(1)   # starts in |0⟩
qc.measure_all()         # add a measurement
qc = transpile(qc, sim)
sim.run(qc, shots=1024).result().get_counts()
```

A fresh qubit starts in **|0⟩**, so measuring it can only ever give 0 — there is no randomness yet.

Result: all 1024 shots read **0**, and none read 1.

Try it: run it again — the counts never change, because |0⟩ has no randomness.

On the Bloch sphere



A fresh qubit sits at the |0⟩ pole, so every measurement returns 0.

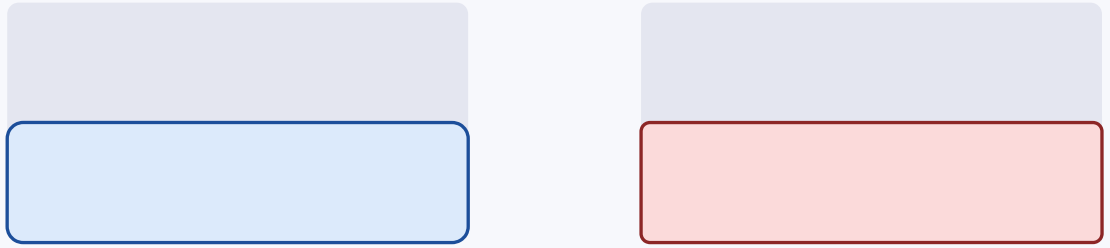
4 Superposition: run it, read the counts

Add one H gate, run 1024 shots, and the qubit's odds show up as counts.

Circuit: H, then measure

$$H|0\rangle = 0.707\ |0\rangle + 0.707\ |1\rangle$$

qc.h(0) makes an equal superposition, so each shot lands on 0 or 1 at random — about half each.



'0' ≈ 512 shots

'1' ≈ 512 shots

≈

Counts are random

Every run differs slightly — say 507 versus 517. That is sampling noise, not a bug.

√N

More shots, tighter odds

The wobble shrinks roughly as 1/√shots: 100 shots swing widely, 100,000 barely move.

#

Fix the seed

Create the simulator with AerSimulator(seed_simulator=42) to get repeatable counts.

5 Predict, run, compare

Say what you expect before every run — then let the counts confirm or correct you.

Circuit	Prediction	Simulator output
Just measure	Always 0	{'0': 1024}
H, then measure	About half 0, half 1	Roughly {'0': 512, '1': 512}
H, H, then measure	Back to 0 — the two paths interfere	{'0': 1024}
X, then measure	Always 1 (X flips 0 to 1)	{'1': 1024}
H + CNOT, then measure	Only 00 or 11 (a Bell pair)	Roughly {'00': 512, '11': 512}

6

Two qubits: build a Bell pair

H then CNOT gives two qubits whose results always agree.

The circuit

$$|\psi\rangle = 0.707\ |00\rangle + 0.707\ |11\rangle$$

Built with qc.h(0) then qc.cx(0, 1): H creates the superposition, CNOT ties qubit 1 to qubit 0.

Measure

After measurement — one outcome

00

with prob. 50%

11

with prob. 50%

01 and 10 never appear — the two results always match.

Perfectly correlated

Each qubit alone looks like a fair coin, yet the pair always agrees — the signature of entanglement.

Check the missing keys

Print the counts: '01' and '10' should be absent entirely on an ideal simulator.

Hardware is noisier

Real devices add errors, so 01 and 10 show up a few percent of the time. The simulator shows the ideal case.

7 How far can a simulator go?

Every extra qubit doubles the memory — a laptop runs out of room near 30 qubits.

A statevector simulator stores one complex number per amplitude · about 16 bytes each

n = 5	n = 10	n = 20	n = 30	n = 40
Amplitudes stored	Amplitudes stored	Amplitudes stored	Amplitudes stored	Amplitudes stored
32 amplitudes	1,024 amplitudes	~1 million amplitudes	~1 billion amplitudes	~1 trillion amplitudes
Memory needed	Memory needed	Memory needed	Memory needed	Memory needed
512 bytes	16 KB	16 MB	16 GB	16 TB

The takeaway for this segment: Today's small circuits run instantly, but memory doubles with every qubit — so simulators become impractical around 30 to 40 qubits.

ONE-LINE TAKEAWAY

A simulator lets you build a circuit, run many shots and read the counts — the fastest way to test every idea in this unit.

Build, run, read the counts

Predict before you run

Small circuits only

→

Next: Your turn — practice exercises

Re-create today's circuits, change the gates, and check that your predictions match the counts.